

CBSE Practice Worksheet

Class 7 | Mathematics | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$
(A) 3:6
(B) 5:8
(C) 15:24
(D) 1:2
2. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$
(A) 2
(B) 8
(C) 15
(D) $\sqrt{15}$
3. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
(A) $x - 2$
(B) $x + 2$
(C) $(x-2)^2$
(D) $x^2 - 4$
4. Simplify: $(a^2 - b^2)/(a + b)$
(A) $a - b$
(B) $a + b$
(C) $a^2 - b$
(D) $a - b^2$
5. What is the cube root of -125?
(A) 5
(B) -5
(C) ± 5
(D) undefined
6. Factorize: $x^2 - 5x - 14$
(A) $(x-7)(x+2)$
(B) $(x+7)(x-2)$
(C) $(x-7)(x-2)$
(D) $(x+7)(x+2)$
7. If $2\blacksquare = 3\blacksquare = 6\blacksquare$, which is true?
(A) $1/z = 1/x + 1/y$
(B) $z = x + y$
(C) $z = xy$
(D) $z = x - y$
8. Solve: $2^{(x+1)} = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$

(D) $x = 6$

9. Solve: $\sqrt{x+7} = x-5$

(A) $x = 9$

(B) $x = 2$

(C) $x = 9$ or 2

(D) No solution

10. Simplify: $(x^3 - 8)/(x - 2)$

(A) $x^2 + 2x + 4$

(B) $x^2 - 2x + 4$

(C) $x^2 + 4$

(D) $x^2 - 4$

Section B: Fill in the Blanks (1 mark each)

11. If $\log x = 2$, then $x = \underline{\hspace{2cm}}$

12. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$

13. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{2cm}}$

14. $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{2cm}}$

15. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{2cm}}$

16. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = \underline{\hspace{2cm}}$

17. $\sqrt[3]{-64} = \underline{\hspace{2cm}}$

18. $\text{HCF} \times \text{LCM}$ of 12 and 18 = $\underline{\hspace{2cm}}$

Section C: Short Answer Questions (2 marks each)

19. Simplify: $(x^2 - 16)/(x^2 + 4)$

20. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

21. If $x = 2 + \sqrt{3}$, find $x + 1/x$

22. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

23. Solve: $\log(x - 1) + \log(x + 1) = 3$

24. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

25. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

--- End of Worksheet ---

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